

SUDAN CONFLICT: A NETWORK APPROACH

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ABSTRACT. In this paper we apply a theory of conflict that explicitly models alliances and enmities in a network framework to study the Sudan conflict. The central assumption is that each group participating in the conflict maximizes the share of resources controlled at the end of the conflict net of the cost of resources invested. We follow König, Rohner, Thoenig, and Zilibotti (2014) to provide a model of a network in conflict. We use this model to derive a structural equation which we then estimate. We build a panel dataset at the fighting group (year, level), covering the period 2003-2010. This period was selected to include the major armed conflict in the Darfur region of Sudan and to exclude the period following the independence of South Sudan in 2011. Our main variables were constructed using data from the Armed Conflict Location and Events Dataset ACLED (2014).

Keywords: Networks, Conflict, Africa, Sudan

JEL Classification: C70, L14, N47.

1. INTRODUCTION

Accounting for alliances and enmities is essential to understand the dynamic of armed conflicts which, in turns, is a necessary condition to achieve nonviolent resolution to these conflicts. Most frequently than not allied groups pursue different goals and compete for the same resources.

In this paper we apply a theory of conflict that explicitly models alliances and enmities in a network framework. The central assumption is that each group participating in the conflict maximizes the share of resources controlled at the end of the conflict net of the cost of resources invested. In this context, the network of allies and enemies imposes externalities over the participants: the success of a group is a function of its *fighting effort* as well as of that of its neighbors in the network. More specifically, the operational performance of a group depends directly on the group's fighting effort as well as on that of its allies (weighted by an allied externality parameter) and inversely on the fighting effort of its enemies (weighted by an enemy externality parameter). In sum, each group's fighting effort produces positive and negative spillover effects that are channeled by the network.

We follow König et al. (2014) and characterize the Nash equilibrium of the game. The fighting effort of each agent is pinned down by its centrality in the network. Each player appropriates a share of the resources on the table that can be expressed as the ratio between a function of the player's degree (i.e., the number of direct enemies and friends) and the sum of all these functions for all the actors in the network. Remarkably, the larger the number of allies, the lower the share of the prize appropriated and the larger the share of the prize net of fighting effort. This is essentially an expression of the free-rider problem. In the same way the network determines the fighting effort and the relative success of each participant, it also determines the size of the conflict. The model predicts that the larger the number of enmity relations, the larger the rent dissipation induced by the conflict (the opposite being true for friendship relations).

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To test the prediction of the model we study an ongoing armed conflict in the territories of the republics of Sudan and South Sudan between 2003 and 2010. Our dataset contains information from two overlapping but differentiated conflicts: the *Second Civil War*, and the *War in the Darfur*. Prior to its division in 2011, Sudan was the largest country in Africa, and the largest Arab country in the world. Throughout its history Sudan has been affected by ethnic conflicts and wars. In particular, the First Civil War (1955–1972), the Second Civil War (1983–2005), and the War in the southern territory of Darfur, are among the deadliest civil conflicts in history. A defining characteristic of this conflict is the large number of local and foreign interest involved. We identify 102 active armed groups displaying alliances and enmities and transitioning from one to the other. These conflicts constitute one of the worst humanitarian tragedies in modern history, a tragedy that each of the authors of this paper deeply regrets. However, the tragedy also provides us with a good source of information to empirically test our model and if for not better reason, to learn from it.

The paper is organized as follows. **Section 2 introduces the model and characterizes the equilibrium effort levels.** Section 3 presents the structural model and discusses the estimation strategy. Section 4 historically contextualizes the conflict and describes the data and the network. Section 5 reports the results. Section 6 concludes.

2. THEORETICAL MODEL

We draw heavily the description of the theoretical model and the characterization of the equilibrium from König et al. (2014).

2.1. The Model Economy. Consider an economy populated by a finite set of players $N = \{1, \dots, n\}$ who are connected in a network $G \in \mathcal{G}^N$, where $\mathcal{G}^N$ denotes the class of networks on the set of nodes N . Any pair of agents $i, j \in N$, $i \neq j$, can be in three possible bilateral states: alliance, neutrality, or hostility. This configuration is represented by the matrix $\mathbf{A} = [a_{ij}]_{1 \leq i, j \leq n}$ where $a_{ij} \in \{-1, 0, 1\}$. Formally, the matrix $\mathbf{A}$ is the *signed adjacency matrix* associated with the network G , where we have that:

$$a_{ij} = \begin{cases} 1 & \text{if } i \text{ and } j \text{ are allied,} \\ -1 & \text{if } i \text{ and } j \text{ are enemies,} \\ 0 & \text{if } i \text{ and } j \text{ are in neutral terms.} \end{cases}$$

Notice that if any pair $i, j \in N$, $i \neq j$ is not in a relationship, we define it to be in neutral terms. Also, as is conventional, the diagonal entries of the matrix $\mathbf{A}$, which correspond to the relationship between agent i with himself, are assumed to be zero.

Define the positive and negative parts of a_{ij} as $a_{ij}^+ = \max\{a_{ij}, 0\}$ and $a_{ij}^- = -\min\{a_{ij}, 0\}$. Then, we have that $a_{ij} = a_{ij}^+ - a_{ij}^-$ for all $i, j \in N$. Hence, we can write $\mathbf{A} = \mathbf{A}^+ - \mathbf{A}^-$ where $\mathbf{A}^+ = [a_{ij}^+]_{1 \leq i, j \leq n}$ and $\mathbf{A}^- = [a_{ij}^-]_{1 \leq i, j \leq n}$. Denote as G^+ and G^- the corresponding subgraphs associated with matrices $\mathbf{A}^+$ and $\mathbf{A}^-$ respectively; thus, G can be written as $G = G^+ \oplus G^-$, where $\oplus$ denotes the graph join. Finally, let $d_i^+ = \sum_{j=1}^n a_{ij}^+$ denote agent i 's number of alliances and respectively $d_i^- = \sum_{j=1}^n a_{ij}^-$ his number of enmities.

To complete the description of the model, we need to specify the payoffs of the agents. Each agent competes for a prize whose value is $V > 0$. Each agent exerts *effort*, which in our case is understood as fighting intensity, in order to win the prize. The share of the prize that each agent receives is determined by a *Contest Success Function* or CSF. The CSF maps the relative fighting

intensity each agent puts in the conflict to the share of the prize he appropriates. Formally, the payoff function for each agent, $\pi_i : \mathcal{G}^N \times \mathbb{R}_+^N \rightarrow \mathbb{R}$ is defined by

$$\pi_i(G, \mathbf{x}) = \frac{\varphi_i(G, \mathbf{x})}{\sum_{j=1}^n \varphi_j(G, \mathbf{x})} V - x_i, \quad (2.1)$$

where $\mathbf{x} \in \mathbb{R}_+^N$ is the vector describing the fighting effort of each of the agents and $\varphi_i : \mathcal{G}^N \times \mathbb{R}_+^N \rightarrow \mathbb{R}$ is agent's i 's *operational performance* (OP). The operational performance of agent i is assumed to depend not only on his fighting effort, but also on the efforts of his allies and enemies. More specifically, we assume the following functional form:

$$\varphi_i(G, \mathbf{x}) = x_i + \beta \sum_{j=1}^n a_{ij}^+ x_j - \gamma \sum_{j=1}^n a_{ij}^- x_j, \quad \beta, \gamma \in [0, 1]. \quad (2.2)$$

2.1.1. Characterization of the equilibrium effort. In this subsection we characterize the **equilibrium fighting effort**. There is **no restriction imposed on the level of effort**, so in principle it could be negative. Note that this is inconsequential for the model, since an effort level of zero is just a matter of normalization.

From the profit functions (2.1) for each agent, the first order conditions yield, for all $i = 1, 2, \dots, n$:

$$\varphi_i = \frac{\left(1 - \frac{1}{V} \sum_{j=1}^n \varphi_j\right) \sum_{j=1}^n \varphi_j}{1 + \beta d_i^+ - \gamma d_i^-}, \quad (2.3)$$

where it is assumed that for all i , the condition $1 + \beta d_i^+ - \gamma d_i^- > 0$ holds, which is a necessary and sufficient condition for the second order conditions to hold for all players. **This** is a system of n equations with n unknowns in which the variables to be solved for are the φ_i 's. To simplify the calculations, we introduce the following notation:

$$\Gamma_i^{\beta, \gamma}(G) = \frac{1}{1 + \beta d_i^+ - \gamma d_i^-} \quad (2.4)$$

$$\Lambda^{\beta, \gamma}(G) = 1 - \frac{1}{\sum_{i=1}^n \Gamma_i^{\beta, \gamma}(G)}. \quad (2.5)$$

Then equation (2.3) can be solved for the **equilibrium operational performance**:

$$\varphi_i^*(G) = \Lambda^{\beta, \gamma}(G) (1 - \Lambda^{\beta, \gamma}(G)) \Gamma_i^{\beta, \gamma}(G) \times V. \quad (2.6)$$

Note that summing over i yields the total economy OP level:

$$\sum_{i=1}^n \varphi_i^* = \Lambda^{\beta, \gamma}(G) \times V, \quad (2.7)$$

which implies that

$$\frac{\varphi_i^*}{\sum_{j=1}^n \varphi_j^*} = \frac{\Gamma_i^{\beta, \gamma}(G)}{\sum_{j=1}^n \Gamma_j^{\beta, \gamma}(G)}. \quad (2.8)$$

The term $\Gamma_i^{\beta, \gamma}$ is a measure of *local hostility level* which captures the positive and negative externalities associated with agent i 's first degree alliances and enemy links. $\Lambda^{\beta, \gamma}$ is a measure of *total OP* in the network. Note that both terms in equations (2.4) and (2.5) are decreasing with β and increasing with γ . Furthermore, note that from (2.7) we can see that the total operational effort is decreasing in the number of alliances and in the parameter β , and on the other hand, it is increasing in the number of enemy links and in the parameter γ . Finally, the payoff expression (2.1) and (2.8) show

that the equilibrium share of the prize awarded to each agent increases in the number of degree one alliances and is decreasing in the number of degree one enmities.

The following proposition (König et al., 2014) characterizes the equilibrium fighting efforts.

Proposition 2.1. *Assume that $\beta + \gamma < 1/\max\{\lambda_{\max}(G^+), d_{\max}^-\}$ and $\beta\lambda_{\max}(A^+) < 1 - \gamma\lambda_{\max}(A^-)$, where $\lambda_{\max}(A^\pm)$ denotes the largest eigenvalue associated with the matrix $A^\pm$. Let $\Gamma^{\beta,\gamma}$ denote the vector whose elements are the $\Gamma_i^{\beta,\gamma}$. Define*

$$\mathbf{c}^{\beta,\gamma}(G) = (\mathbf{I}_n + \beta\mathbf{A}^+ - \gamma\mathbf{A}^-)^{-1}\Gamma^{\beta,\gamma}(G), \quad (2.9)$$

which is a vector whose elements, $c_i^{\beta,\gamma}(G)$, describe the centrality of agent i in the network. Then, there is a unique Nash equilibrium with effort levels given by

$$\mathbf{x}_i^*(G) = \Lambda^{\beta,\gamma}(G)(1 - \Lambda^{\beta,\gamma}(G))c_i^{\beta,\gamma}(G) \times V, \quad (2.10)$$

for all $i \in N$. The equilibrium payoffs are given by

$$\pi_i^*(G) = V(1 - \Lambda^{\beta,\gamma}(G))(\Gamma_i^{\beta,\gamma}(G) - \Lambda^{\beta,\gamma}(G)c_i^{\beta,\gamma}(G)) \quad (2.11)$$

Note how the centrality measure $c_i^{\beta,\gamma}(G)$ influences the outcome of the game; the equilibrium fighting effort is proportional to the player's centrality in the network. It is noteworthy that the relative fighting effort between two agents only depends on their relative centrality:

$$\frac{x_i^*(G)}{x_j^*(G)} = \frac{c_i^{\beta,\gamma}(G)}{c_j^{\beta,\gamma}(G)}.$$

This result is consistent with other general results about equilibrium strategies in network games with complementarities (cf. Ballester, Calvó-Armengol, and Zenou, 2006; Liu, Patacchini, Zenou, and Lee, 2012).

2.1.2. Heterogeneous fighting efforts. When we characterized the equilibrium fighting efforts in section 2.1.1 it was implicitly assumed that all agents have access to the same homogeneous fighting technology to transform effort into fighting intensity. Therefore, the characterization of the equilibrium efforts focused only on the structure of the network as the only source of heterogeneity. In this section, we develop an extension that will help us introduce other sources of heterogeneity. This is important because in reality, military groups differ in their strength, wealth, size, access to weapons, etc. Furthermore, this extension will be important in order to derive the structural equation that will be estimated.

Assume that agent's i operational performance can be written as

$$\varphi_i(G, \mathbf{x}) = \tilde{\varphi}_i + x_i + \beta \sum_{j=1}^n a_{ij}^+ x_j - \gamma \sum_{j=1}^n a_{ij}^- x_j, \quad (2.12)$$

where $\tilde{\varphi}_i$ is an additive term which enters agent i 's operational performance. Hence, the payoff equation from equation (2.1) can be modified as follows:

$$\pi_i(G, \mathbf{x}) = V \frac{\tilde{\varphi}_i + x_i + \beta \sum_{j=1}^n a_{ij}^+ x_j - \gamma \sum_{j=1}^n a_{ij}^- x_j}{\sum_{j=1}^n (\tilde{\varphi}_j + x_j + \beta \sum_{k=1}^n a_{jk}^+ x_k - \gamma \sum_{k=1}^n a_{jk}^- x_k)} - x_i. \quad (2.13)$$

It is noteworthy that even with heterogeneous fighting technologies, the equilibrium OP levels are still given by equation (2.6). Furthermore, the equilibrium share of the prize is also given by equation (2.8). However, the equilibrium fighting efforts change due to $\tilde{\varphi}_i$. They are characterized in the next proposition (König et al., 2014).

Proposition 2.2. Define $\tilde{\varphi}$ as the vector whose components are the $\tilde{\varphi}_i$'s. Under the same assumptions as in Proposition 2.1 with the additional condition

$$V \times \Lambda^{\beta, \gamma}(G)(1 - \Lambda^{\beta, \gamma}(G))\Gamma^{\beta, \gamma}(G) > \tilde{\varphi},$$

the equilibrium fighting efforts are given by

$$\mathbf{x}^* = (\mathbf{I}_n + \beta \mathbf{A}^+ - \gamma \mathbf{A}^-)^{-1} (V \times \Lambda^{\beta, \gamma}(G)(1 - \Lambda^{\beta, \gamma}(G))\Gamma^{\beta, \gamma}(G) - \tilde{\varphi}). \quad (2.14)$$

Equation (2.14) will permit us to introduce observable and unobservable sources of heterogeneity in the structural model to be estimated.

3. STRUCTURAL MODEL AND ESTIMATION STRATEGY

In this section, we present the structural model that we estimate. We focus on a conflict that repeats itself over several years which is embedded in a network of enmities and alliances. We make the assumption that agents are not playing a repeated game but a one-shot game. This simplifying assumption is to abstract away from reputation and repeated game effects. Furthermore, we consider that the network is stable and the variation in the outcome is due to individual specific shocks. The inclusion of this individual specific shocks is necessary not only to make the model more realistic and in accordance to real data, but also to provide identification of the parameters to be estimated.

For our specification, we specialize the model of heterogeneous fighting technologies proposed in section 2.1.2. Consider the operational effort given by equation (2.12). We impose a functional form into the heterogeneous shock $\tilde{\varphi}_i$ in order to provide for the source of observable and unobservable heterogeneity. Formally, consider the linearly additive form

$$\tilde{\varphi}_{it} = \mathbf{z}'_{it} \boldsymbol{\alpha} + e_i + \varepsilon_{it}, \quad (3.1)$$

where $\mathbf{z}'_{it}$ is a vector of observable shifters with coefficients $\boldsymbol{\alpha}$, e_i is a time-invariant group specific unobservable shifter and finally, ε_{it} is an iid, zero-mean unobservable shifter.

Therefore, we can write a group's OP from equation (2.12) as follows:

$$\varphi_{it}(G, \mathbf{x}) = x_i + \left[\beta \sum_{j=1}^n a_{ij}^+ x_{jt} - \gamma \sum_{j=1}^n a_{ij}^- x_{jt} \right] + \mathbf{z}'_{it} \boldsymbol{\alpha} + e_i + \varepsilon_{it}. \quad (3.2)$$

The equilibrium characterizations then follows proposition 2.2. In particular, note that as previously discussed, total OP in the economy is independent of $\tilde{\varphi}_{it}$:

$$\sum_{i=1}^n \varphi_{it}^* = \left(1 - \frac{1}{\sum_{i=1}^n \frac{1}{1 + \beta d_i^+ - \gamma d_i^-}} \right) \times V. \quad (3.3)$$

Therefore, total OP in the economy is fully characterized by the structure $(\beta, \gamma, (d_i^+, d_i^-)_{i=1}^n)$, which incorporates time-invariant parameters of the OP equation and the network structure. Furthermore, this structure is independent of the individual shocks $(\mathbf{z}_{it}, e_i, \varepsilon_{it})_{i=1}^n$. This is remarkable, because it simplifies in substantially the estimation procedure we need to use because there is no aggregate level feedback of unobserved heterogeneity on the agents' individual efforts.

Finally, combining the individual level of effort given by equation (2.14), our specification of OP given by equation (3.2) and the equilibrium condition given by equation (3.3) we obtain the

individual equilibrium effort levels, which will be the structural equation that we will estimate:

$$x_{it}^* = -\beta \sum_{j=1}^n a_{ij}^+ x_{jt}^* + \gamma \sum_{j=1}^n a_{ij}^- x_{jt}^* - \mathbf{z}_{it}' \boldsymbol{\alpha} + u_i - \varepsilon_{it}, \quad (3.4)$$

where the time-invariant term of unobserved individual heterogeneity has the form

$$u_i = e_i + \Lambda^{\beta, \gamma}(G)(1 - \Lambda^{\beta, \gamma}(G))\Gamma_i^{\beta, \gamma}(G) \times V. \quad (3.5)$$

Equation (3.4) is our structural model. We estimate this equation using standard panel-data methods. In particular, given the time invariant term u_i described by equation (3.5), we use a fixed-effects specification.

In order to estimate the parameters of interest in equation (3.4) we need to first discuss some identification issues. We follow the estimation strategy proposed in Bramoullé, Djebbari, and Fortin (2009). Firstly, note that in the structural equation the time-invariant unobserved heterogeneity term u_i can be potentially correlated with neighboring fighting efforts x_{jt}^* through the term $\Lambda^{\beta, \gamma}(G)$. Recall that this term is a measure of total operational performance in the network. Intuitively, any group in the network will tend to increase or decrease its fighting effort because they face a similar environment as their neighbors, be they friends or allies. The literature on peer effects and spillovers in networks (see Manski, 1993; Bramoullé et al., 2009) calls this problem *correlated effects*. Our panel data structure allows us to control for these correlated effects.

Consistent estimation of the parameters of interest β and γ requires that we have exogenous sources of variations in allies and enemies fighting efforts which are different from determinants of a group's own fighting effort. Our modeling assumption of the heterogeneous fighting technologies given by equation (3.1) allows us to exclude exogenous observable shifters in neighbor's fighting efforts from the determination of x_{it}^* , i.e., they only effect on x_{it}^* is indirectly through the observable fighting efforts of the allies and neighbors x_{jt}^* . This particular identification problem of assessing how the average behavior of agents in the network affects individual behavior is called the *reflection problem* in the literature (Manski, 1993). Hence, we can exploit the exogenous shifters z_{jt} as instruments for $\sum_{j=1}^n a_{ij}^\pm x_{jt}^*$ while controlling for the shifters of z_{it} . Furthermore, note also that our assumption on the form of the heterogeneous fighting technologies allows us to exclude the covariates of neighbors of neighbors from the structural equation, since they do not appear in it, in contrast with the existing literature (cf. Bramoullé et al., 2009).

As it turns out, finding strong instruments that satisfy the exogeneity restriction is not an easy job. Any time-invariant candidate for an instrumental variable would be absorbed by the fixed-effect term. On the other hand, time varying shocks occurring at country level would also violate the exogeneity restriction by affecting everyone in the network. Therefore, that would rule out natural experiments as an identification strategy.

Following König et al. (2014) who in turn follows Miguel, Satyanath, and Sergenti (2004) and Eynde (2011), we use time-varying climatic shocks, i.e. rainfall, affecting a fighting group area of operation. The rationale for choosing rainfall is that high precipitations levels are associated with higher agricultural surplus that would increase reservation wages for laborers and would lead to an increase in the opportunity cost of fighting. This argument seems plausible given that Sudan is a highly agricultural country (Nashashibi, 1980). On the other hand, high precipitation levels have the effect of making military logistics more complex, for example, by increasing the cost of troop transportation and upkeep of a fighting unit. (see Rogall and Guariso, 2013)

4. EMPIRICAL APPLICATION—THE SUDAN WAR

4.1. Historical Context. In this application we study the armed conflict in Sudan between 2003 and 2010. Our dataset contains information from two overlapping but differentiated conflicts: the *Second Civil War*, and the *War in the Darfur*. For the purpose of our analysis we call Sudan to the territory that now comprises the Republic of the Sudan and the Republic of South Sudan.¹ In this subsection we briefly contextualize the origins of the armed conflict in Sudan.

Sudan is the largest country in Africa in terms of area, and the largest Arab country in the world. Sudan's population was calculated at 25 million according to the 1993 census and it is currently estimated by the CIA at 45 million. Throughout its history Sudan has been affected by ethnic conflicts and wars (Daly and Sikainga, 1993). Among these wars, the First Civil War (1955–1972), the Second Civil War (1983–2005), and the War in the southern territory of Darfur, an ongoing conflict initiated in 2003, are among the deadliest civil conflicts in history.

The transitional institutions created in Sudan in the aftermath of the independence from Egyptian-British control in 1953 left the civil service and administration on the hands of Northern Sudanese people who largely excising the Southern Sudanese from the government. This asymmetry in the distribution of power between North and South exacerbated the preexisting tensions between these regions. Broken promises by the Arab-led Khartoum government to create a federal system led to a mutiny by Southern troops in the Equatoria Province on August 18th, 1955. What originally started as a low-intensity civil war aimed at obtaining regional autonomy and representation turned into a full scale civil war that lasted seventeen years and causing half million deaths. The war would eventually come to an end (or at least to a halt) with the signing of the Addis Ababa accords on March 27, 1972. These agreements guaranteed autonomy for a southern region composed of the provinces of Equatoria (today Al Istiwai), Bahr al Ghazal, and Upper Nile (today Aali an Nil).²

By the end of 1983, the northern government in Khartoum had advanced in its campaign to promote Islam by incorporating traditional Islamic punishments drawn from Shari'a (Islamic Law) into the penal code. These new rules were also imposed on Southerners and other non-Muslims living in the north who would not wait to rise up reactivating a long lasting conflict between North and South. The Second Civil War is seen by many as the continuation of the first war (Johnson, 2003) .

The period from 1984 to 2003 is characterized by the failure of any attempt at peace negotiations. On the military ground, the conflict between the central Sudanese government and the southern Sudan People's Liberation Army (SPLA) (now the official army of South Sudan) left behind two million deaths and more than four million displaced. The number of civilian casualties is estimated to be the highest of any war since World War II. This period is also marked by a large number of human rights violations, including mass killings, ethnic cleansing, and slavery.

In the core of this conflict are the demands for independence, the abolition of the Islamic Law, the abolition of military agreements with Egypt and Libya, and the control over the natural resources (oil) discovered by the end of the 70's in what is now the territory of South Sudan. During most of the conflict, the SPLA maintained the control of large areas of Equatoria, Bahr al Ghazal, and Upper Nile provinces operating additionally in the southern portions of Darfur, Kordofan, and Blue Nile provinces. At the same time, the Central Government controlled a number of the major southern towns and cities, including Juba, Wau, and Malakal.

¹The Republic of South Sudan gained its independence from Sudan in 2011 leading to the division of the territory of Sudan into the two republics mentioned above.

²For a detailed analysis of Sudan's civil wars see Johnson (2003).

The little success of the peace negotiations can be explained partially by the instability of the central government in Sudan. The ruling regime was usually challenged by the military or by opposing political interests. At the same time, the conflict proved to be highly fragmented with respect to interests and organization of the participating armed groups. For the period considered in our empirical application we identified 102 active groups which displayed some alliances over the course of the eight years in our sample, but mostly engaged in violent attacks against each other. This fragmentation is also apparent in the number of foreign interest participating in the conflict where we find armed groups from Uganda, Chad, Eritrea, Ethiopia, and the U.N.

Most of the groups on either side suffered from internal divisions giving rise to competing factions that would eventually turn violent against each other. For example, the Global Security Organization, reports that in August 1991, internal dissention among the rebels led opponents of Colonel Garang's leadership of the SPLA to form the so-called Nasir faction of the rebel army. In September 1992, William Nyuon Bany formed a second rebel faction, and in February 1993, Kerubino Kwanyin Bol formed a third rebel faction. On April 5, 1993, the three dissident rebel factions announced a coalition of their groups called SPLA United at a press conference in Nairobi, Kenya. After 1991, the factions clashed occasionally and thus, the rebels lost much of their credibility with the West. Some of these divisions in the southern forces were incentivized by the central government through a manipulative effort of highlighting ethnic differences among these forces. These ethnical differences are also captured in our sample where many of the armed groups are ethnically identified. The same type of fragmentation was observed within the forces loyal to the rulers in Khartoum. Western and eastern regions of Sudan started to perceive the Arab-dominated government as alienating and non responsive to the needs of the population. A fraction of the population in these regions grow sympathetic to the southern rebels led by the Sudan People's Liberation Movement/Army (SPLM/A), and in some cases, prompted them to flight alongside it.

In July 2002, the Government of Sudan and the SPLM/A agreed to the Machakos Protocol which among other things acknowledged the right of southern Sudan to self-determination. Talks and deliberations continued during the next three years leading to the signature of a peace agreement in Nairobi on January 9th, 2005. This agreement is sometimes taken to be the end of the conflict and prepared the terrain for the future referendum leading to the Independence of South Sudan in 2011.

Our sample includes incidents taking place within the ongoing conflict know as The War in Darfur.³ Originated in the Darfur region of Sudan, it began in February 2003 after the Sudan Liberation Movement/Army (SLM/A) and Justice and Equality Movement (JEM) rebel groups rose against the government of Sudan. The rising followed up allegations indicating decades of political oppression against the non-Arab population, and economic neglect by the Sudanese government. In turn, the government moved to crush the rebellion by force implementing ethnic cleansing against Darfur's non-Arabs. One side of the conflict was composed mainly of Sudanese military, the Sudanese Police Forces, and the Janjaweed, a Sudanese militia group (supported by the Sudanese government) recruited mostly among Arabized indigenous Africans and a small number of Bedouin of the northern Rizeigat. On the other side, the most notorious groups were the SLM/A and the JEM, recruited primarily from the non-Arab Muslim Fur, Zaghawa, and Masalit ethnic groups (De Waal, 2004).

Degomme and Guha-Sapir (2010) reports different studies assessing the number of excess deaths in Darfur between 2003 and 2008. The paper estimates that the number of excess deaths is 298, 271

³For a discussion on the War of Darfur see Johnson (2003).

with a 95%CI ranging from 178,258 to 461,520. Moreover, they report estimates indicating that the number of displaced individuals by 2008 was around 2 million.

4.2. Data. We build a panel dataset at the fighting group (year, level), covering the period 2003-2010. This period was selected to include the major armed conflict in the Darfur region of Sudan and to exclude the period following the independence of South Sudan in 2011.

Our main variables were constructed using data from the Armed Conflict Location and Events Dataset ACLED (2014). However, we relied on additional external datasets to construct some covariates and the Instrumental Variables. Below we make a brief description of each them.

Fighting efforts: We measure fighting efforts using data from the ACLED, 2014. This dataset contains 581 geolocalized violent events taking place in Sudan among 102 fighting groups over the 2003-2010 period. For each violent event, information is provided on the exact location, the date and the identities of the fighting groups involved in the event. To construct our main variable of fighting effort we take the sum over all fighting events against other armed groups in which a given fighting group is involved during a year.

Alliances and enmities: Unique to the ACLED data is dyadic information on the composition of each opposing side involved in a given event. The two main variables of this dataset provide us with direct measures of our main theoretical variables, x_t (the vector of equilibrium fighting efforts in year t), and the adjacency matrices of alliances and enmities observed on the battlefield. We measure x_{it} , the fighting effort of a group i in year t , as the total number of ACLED violent events the group participates to.

Controls and Instruments: The following variables are used to generate the set of standard control variables and of Instrumental variables (IVs).

- **Foreign:** This dummy variable takes a value of 1 for all foreign actors, and a value of 0 of all domestic fighting groups.
- **Fighting Effort Outside the DRC:** For all groups we also compute the total number of fighting events in which they are involved outside Sudan.
- **Government Organization:** This variable takes a value of 1 for fighting groups that are official government organizations of one of the countries involved, and 0 otherwise.
- **Rain:** We construct, for each fighting group, the geolocalised “barycenter” containing all the ACLED fighting events involving the group. This gives as a measure of the spatial zone of operation of the group. Then we use the rainfall data from Schneider et al. (2011) to compute the average rainfall for each year in the zones of operation.

4.3. Network of Fighting Groups in Sudan. We code two groups i, j as being allied (i.e.: $a_{ij}^+ = a_{ji}^+ = 1$) if they fought alongside in at least one event over the whole sample period, and, in addition, they have never fought on opposing sides in any event. Similarly, we code two groups as being enemy (i.e.: $a_{ij}^- = a_{ji}^- = 1$) if they have fought on at least one occasion on opposing sides during the sample period, and have never been allied in any event. We code all other dyads as neutral (i.e.: $a_{ij}^+ = a_{ij}^- = 0$).

The summary statistics of our variables are displayed in Table 1.

Our data allows us to identify 143 dyads. Out of these 143 dyads, 132 represent enmity relationships and 11 represent friendship relationships. Table 2 reports descriptive network statistics. Our network exhibits low average degree, approximately 75% of the actors in the network engaged with less than two other actors.

Not surprisingly, our network also exhibits high clustering. For this, we ordered nodes with respect to their degree and identify the most relevant actors to be the Military Forces of Sudan, the

TABLE 1. Descriptive Statistics

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Median	Pctl(75)	Max
Fight. Eff. Allies	816	2.59	11.14	0	0	0	0	123
Fight. Eff. Enemies	816	33.97	53.18	0	0	4	48.2	331
Rain Ally	816	0.58	1.53	0.00	0.00	0.00	0.00	10.54
Rain Enemy	816	11.66	23.08	0.31	2.94	4.47	9.71	236.19
Fight. Effort	816	2.76	10.41	0	0	0	1	123
Foreign Effort	816	0.06	0.29	0	0	0	0	3
Government	816	0.06	0.24	0	0	0	0	1

TABLE 2. Network Statistics

Average Degree	2.804
Network Diameter	9
Average Path length	3.323
Graph Density	0.028
Connected components	13
Avg. Clustering Coefficient	0.373
Avg. Path Length	3.324

Military Forces of Southern Sudan, the Sudan Liberation Movement/Army, the Lord's Resistance Army, and the Janjaweed. It is interesting to observe that this selection of actors is also consistent with the most relevant actors identified in the previous section. Table 3 reports descriptive node statistics for each of these military groups. Figure 1 depicts the complete conflict network, the network of friendship, the network of enmity, and the network of the five players with the highest degree.

TABLE 3. Main Actor Node Statistics

Actor	Degree	Eccentricity	Closeness Centrality	Betweenness Centrality	Clustering Coefficient	Number of triangles	Eigenvector Centrality
Military Forces of Sudan (1989-)	22	6	2.23	1014.59	0.10	22.00	1.00
Military Forces of Southern Sudan (2005-2011)	19	5	2.21	1228.22	0.04	7.00	0.59
SLM/A: Sudan Liberation Movement/Army	15	7	2.55	310.37	0.18	19.00	0.84
LRA: Lord's Resistance Army	13	6	2.30	604.51	0.13	10.00	0.61
Janjaweed	9	7	2.84	67.45	0.28	10.00	0.54

An alternative visualization of the network for the most relevant players is reported in the appendix.

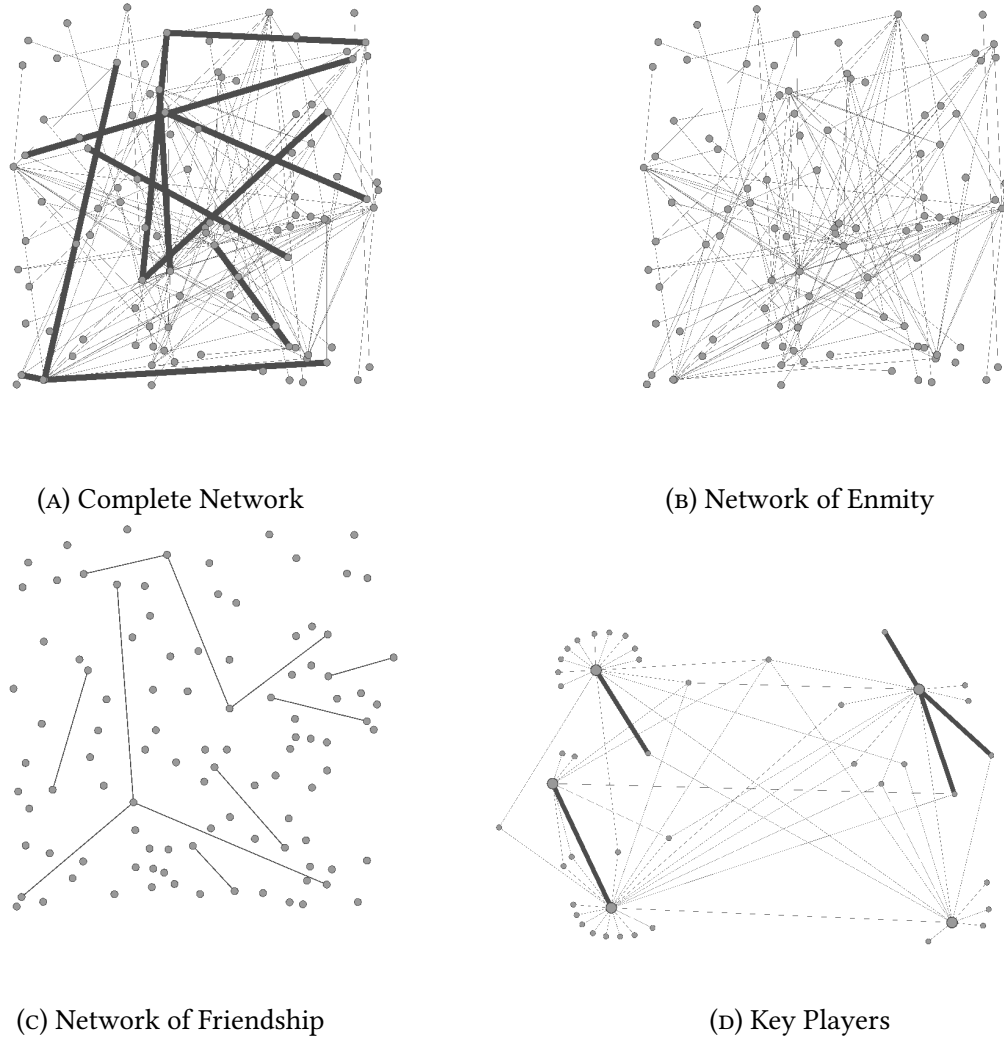


FIGURE 1. Network Structure

5. ESTIMATES OF THE FIGHTING EXTERNALITIES

We now present the estimates of the structural equation (3.4) based on a panel dataset containing 102 armed groups between 2003 and 2010.

Table 4 reports the results from the baseline specifications. In Column (1) we report an OLS specification including group fixed effects and year dummies. The total fighting of allies decreases the group's fighting effort, whereas the total fighting of enemies increases its fighting effort. These results are in line with the theory exposed, although the coefficient of total fighting of enemies is not relevant.

In Column (2) we control for the current and lagged rainfall in the barycenter of the action zone of groups, allowing for linear and quadratic effects. This specification follows the one used by König et al. (2014). The results are similar to the OLS model in terms of interpretation. However the magnitude of both estimates have changed considerably, whereas the total fighting of enemies became statistically not significant.

Finally, Column (3) reports the result from the second stage of a 2SLS specification including lagged weather shocks and Foreign Effort as a control variable. In this specification, total fighting

of enemies and total fighting of allies are instrumented by the one-period lag of average rainfall in enemies and allies action's zones (both a linear and square terms are used). The coefficients of the variables of interest have the expected sign. Again, only total fighting of enemies is significant.

TABLE 4. Baseline Regressions

	Fighting Efforts		
	(1)	(2)	(3)
Fight. Eff. Allies	-0.23*** (0.03)	-0.54*** (0.20)	-0.45** (0.20)
Fight. Eff. Enemies	0.02* (0.01)	0.13 (0.09)	0.14 (0.09)
Group FE, Time Dummies	Yes	Yes	Yes
Control for Rainfall	No	Yes	Yes
Additional Controls	No	No	Yes
Estimator	OLS	IV	IV
Observations	816	714	714
R ²	0.09	0.07	0.07
Adjusted R ²	0.07	0.06	0.06

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Standard treatments of instrumental variables regression stress the idea that for instruments to be valid they must be exogenous. It is also important that the “relevance” condition for a valid instrument holds as well. If the instruments are “weak”, then bias can be even more pronounced than in the OLS case. In this sense, it is important to assess the strength of our instruments.

Following Stock and Yogo (2005) methodology, we run the first stage regression to assess its F -statistic and then compare it to the thresholds of instruments’ validity proposed by the authors when there are more than one endogenous regressors and multiple instruments.

Table 5 below reports the F -statistics of the first stage of the endogenous regressors “Fighting Efforts Allies” (Column 1) and “Fighting Efforts Enemies” (Column 2). By proceeding with the inspection⁴ we conclude that our statistics do not satisfactorily reach such thresholds. Instruments for “Fighting Efforts Allies” are only valid at the significance level of 30%, whereas Instruments for “Fighting Efforts Enemies” are valid at 10%. Thus we conclude that the results displayed in Table 5 suggest that our instruments are weak. As we already argued, one consequence of weak instruments is that IV estimators are in general biased.

As explained in Section 3 we use time-varying climatic shocks, i.e. rainfall, affecting a fighting group area of operation as instruments. At a first glance, it is suggestive that this variable may capture the exogenous variation of our endogenous regressors. However, it also seems plausible to argue that Rainfall in Enemies action zone, for instance, may influence some of its opponents’ attack decision. If this argument is valid, then we can conclude that our instruments are problematic and we should look for extra source of exogenous variability to deal with the endogeneity we have in our two main regressors.

⁴Table 5.1 in Stock and Yogo (2005) reports the critical values for the weak instrument test based on TSLS bias. For the case in which there are 2 endogenous regressors and 8 instrumental variables, the thresholds for validity and its critical levels are: 17.70 (5%), 10.22 (10%), 6.20 (20%), 4.73 (30%).

TABLE 5. First Stage—IVs

	Fighting Efforts	
	Allies (1)	Enemies (2)
Foreign Effort	3.70*** (1.15)	5.23 (4.35)
y05	0.15 (0.91)	7.22** (3.45)
y06	-2.63*** (0.96)	-17.10*** (3.62)
y07	-0.76 (0.91)	3.76 (3.46)
y08	-2.33*** (0.88)	-15.71*** (3.32)
y09	-2.38*** (0.90)	-22.50*** (3.42)
y10	-2.59*** (0.93)	-19.20*** (3.54)
Rain Ally	-2.82** (1.26)	-5.57 (4.76)
Rain Ally Sqr.	0.33*** (0.13)	0.20 (0.47)
Rain Enemy	0.08 (0.09)	-0.17 (0.36)
Rain Enemy Sqr.	-0.001* (0.0004)	0.0001 (0.001)
Rain Ally ($t - 1$)	-3.42*** (1.28)	1.21 (4.86)
Rain Ally ($t - 1$) Sqr.	0.32** (0.13)	0.06 (0.49)
Rain Enemy ($t - 1$)	0.18* (0.10)	0.41 (0.38)
Rain Enemy ($t - 1$) Sqr.	0.0004 (0.0004)	0.0001 (0.001)
Observations	714	714
R ²	0.11	0.23
Adjusted R ²	0.09	0.19
F Statistic (df = 15; 597)	4.94***	11.93***

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$
Using Fixed Effects Model

6. CONCLUSIONS

Sudan is the largest country in Africa in terms of area, and the largest Arab country in the world. Sudan's population was calculated at 25 million according to the 1993 census and it is currently estimated by the CIA at 45 million. Throughout its history Sudan has been affected by ethnic conflicts and wars. Among these wars, the First Civil War (1955–1972), the Second Civil War (1983–2005), and the War in the southern territory of Darfur, an ongoing conflict initiated in 2003, are among the deadliest civil conflicts in history.

Based on the work of König et al. (2014), we presented in this paper a theoretical model for the Sudan conflict in the period of 2003–2010. In this framework, different groups compete for the appropriation of a prize (with value normalized to one). For that, we introduced a network of alliances and enmities which generates externalities to the incentives of fighting of each actor.

Using the structural equation which characterizes the Nash Equilibrium of this contest, we estimate these externality parameters. In general, the signs of these parameters are in line with the theory exposed in this work, although the coefficient of total fighting of enemies shows either lack of relevance (OLS specification) or lack of statistical significance (IV specifications).

Following the Stock and Yogo (2005) methodology, we ran the first stage regression to access its F -statistic and then compared it to the thresholds of instruments' validity proposed by the authors when there are more than one endogenous regressors and multiple instruments. We conclude that the results displayed in Table 5 suggested that our instruments are weak. Looking for extra source of exogenous variability to deal with the endogeneity we have in our two main regressors seems the most reasonable way to revert such problem.

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APPENDIX A. ACTORS

TABLE 6. Armed Groups in Sudan Conflict

	Armed Groups
1	African Union Forces (2002-)
2	Agaar Ethnic Militia
3	Al-Falata Ethnic Militia
4	Al-Sa'ada Ethnic Militia
5	Ambororo Ethnic Militia
6	Arab Ethnic Militia
7	Arrow Boys Militia
8	Awan Ethnic Militia
9	Awlad Omran Ethnic Militia
10	Bali Ethnic Militia
11	Bari Ethnic Militia
12	BC: Beja Congress
13	Beni Amir Ethnic Militia
14	Birgid Ethnic Militia
15	Bongo Ethnic Group
16	Bul Ethnic Militia
17	Burgo Ethnic Militia
18	Dar Na'ila Ethnic Militia
19	Dinka Agar Ethnic Militia
20	Dinka Ethnic Militia
21	Dinka Joth Ethnic Militia
22	Dinka Luach Ethnic Militia
23	Dinka Ngok Ethnic Militia
24	Dinka Nyang Ethnic Militia
25	Dinka Pakam Ethnic Militia
26	Dinka Rek Ethnic Militia
27	Eastern Front
28	Equatoria Defence Force
29	EUFOR Chad/CAR: Military Operation in Eastern Chad and North Eastern Central African Republic
30	Former SPLA/M: Sudanese People's Liberation Army/Movement
31	Gatlauk Gai Rebel Group
32	George Athor Deng Rebel Group
33	Gholfan Ethnic Militia
34	Gimir Ethnic Militia
35	Gok Ethnic Militia
36	Habbaniyah Ethnic Militia
37	Hawazmah Ethnic Militia
38	Hotiya-Baggara Ethnic Militia (Somalia)
39	Janjaweed
40	JEM: Justice and Equality Movement
41	JEM-UFN: Justice and Equality Movement (Unity Faction Nur)
42	Jie Ethnic Militia
43	Jikany Ethnic Militia
44	Jul-bel Ethnic Militia (South Sudan)
45	Jur Ethnic Militia
46	Lango Communal Militia
47	Leek Ethnic Militia (South Sudan)
48	LJM: Liberation and Justice Movement
49	Lou Nuer Ethnic Militia
50	Lou Nuer Ethnic Militia (South Sudan)
51	LRA: Lord's Resistance Army

TABLE 6 (CONT).. Armed Groups in Sudan Conflict

	Armed.Groups
52	Luach Ethnic Militia
53	Mahadi Ethnic Militia
54	Masalit Ethnic Militia
55	Military Forces of Chad (1990-)
56	Military Forces of Ethiopia (1991-)
57	Military Forces of Southern Sudan (2005-2011)
58	Military Forces of Sudan (1989-)
59	Military Forces of Uganda (1986-)
60	Militia (Gabriel Tang)
61	Misseriya Ethnic Militia
62	Moru Ethnic Militia
63	Mule Ethnic Militia (South Sudan)
64	Mundari Ethnic Militia
65	Murahaleen Ethnic Militia
66	Murle Ethnic Militia (South Sudan)
67	Newiba-Aballa Ethnic Militia
68	NMRD: National Movement for Reform and Development
69	Norab Ethnic Militia
70	NRF: National Redemption Front
71	Nuba Ethnic Militia
72	Nuer Ethnic Militia
73	Nuwayba Ethnic Militia
74	ONLF: Ogaden National Liberation Front
75	PDF: Popular Defence Forces
76	Police Forces of Southern Sudan (2005-2011)
77	Police Forces of Sudan (1989-)
78	Rizaygat Ethnic Militia
79	Salamat Ethnic Militia
80	Shilluk Ethnic Group
81	Shilluk Ethnic Militia
82	SLM/A: Sudan Liberation Movement/Army
83	SLM/A-FW: Sudan Liberation Movement/Army-Free Will
84	SLM/A-Minnawi: Sudan Liberation Movement/Army (Minnawi Faction)
85	SLM/A-Nur: Sudan Liberation Movement/Army (Nur Faction)
86	SLM/A-Shafi: Sudan Liberation Movement/Army (Ahmed Abdel Shafi Faction)
87	SLM/A-Unity: Sudan Liberation Movement/Army (Unity Faction)
88	SLM/A-Wahid: Sudan Liberation Movement/Army (Abdul Wahid Faction)
89	SPLA/M-Joint Integrated Units: Sudanese People's Liberation Army/Movement Joint Integrated Units
90	SSDF: South Sudan Defence Forces
91	Tepeth Ethnic Militia
92	Terjem Ethnic Militia
93	Toposa Ethnic Militia
94	UFDD: Union of Forces for Democracy and Development
95	UFR: Union of Forces for Resistance
96	UNAMID: United Nations/African Union Hybrid Operation in Darfur
97	UNMIS: United Nations Mission in Sudan
98	URFF: United Revolutionary Forces Front
99	Western Equatoria Community Defence Force
100	Wildlife Protection Personnel
101	Zaghawa Ethnic Militia
102	Zande Ethnic Militia

APPENDIX B. LARGEST ACTORS

